

Thermal Mining

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Thermal Mining I

- *Mechanical Machines have dominated rock cutting technology with out blasting.
- *Thermal fragmentation system holds promise for Rock having compressive strength more than 140 Mpa.
- *Available rock Cutting Mechanical machine can not cut rock having compressive strength 140Mp and above.
- *Present technology of Machine Rock cutting is by exerting high compressive force on rock.
- *This result the breakage of cutting teeth and machine.

Thermal Mining II

- *Only method of breaking the Rock is by fracturing Rock in tension.
- *Thermal method proves to be an option of breaking Rock with out blasting.
- *Thermal option can be considered, where blasting is not possible.

Heating Rock I

- *Historically hard rock is broken by heating with fire on rock.
- *Then cold water is poured on hot rock.
- *Sudden changes in temperature of the rock makes it to crack with out blasting.
- *Braking rock by heating externally is not very efficient method.
- *Application of very high temperature on rock surface does not yield result.

Heating Rock II

- *Poor thermal conductivity of rock makes surface melt, without heating the interior.
- *Surface heating is not an option for thermal breaking of rock.
- *Melting point of silica is 1710 degree centigrade.

Heating by Microwave I

- *Microwave heating can fragment hard rock.
- *It can also weaken rock to improve performance of other mining method or secondary breaking.
- *Micro wave can penetrate rock.
- *Electro magnetic energy is converted by rapid vibration of molecules at a location (hot Spot inside the rock).
- *Hot rock then expands according to its specific thermal expansion.
- *Surrounding cool rock does not expand.

Heating by Microwave II

- *Swelling hot spot inside rock causes tensile stress in rock.
- *Tensile stress in surrounding rock, eventually fragmenting the rock.
- *Rock can also be heated with lower energy to induce just enough fracture to make it easy for mechanical method such as ripper can complete the excavation.
- *Microwave generator at frequency 2.45GHz with power rating at 25kw fracture large rock in laboratory. (100 kg boulder).

Heating by Microwave III

*Time required to break 80kg iron ore (Magnaniferous) is 180 second.

*Microwave penetration is expressed by following formula

$$d = uEaT/(1-P)$$

d = stress, stress is independent of the size of inclusion

u = constant related to shape. Range 1 to 2/3

E = Young's modulus

a = expansion coefficient

T = temperature raise

P = poisson's ratio

Heating by Microwave IV

- *Ability of microwave to fracture rock does not depend on compressive strength.
- *Ability to fracture rock grows up as the temperature goes up to a certain maximum temperature.
- *Internal temperature at which stress are maximum, vary from rock to rock.
- >>Critical temperature for Basalt is 500 degree C
- >>Critical temperature for Granite is 570 degree C

Heating by Microwave V

- *At higher temperature, young modules fails and rock softens.
- *At higher temperature rock does not crack as much as at critical temperature.

Use of Microwave I

- *A more energy efficient application of micro wave is softening hard rock in advance of mechanical excavation.
- *Rock is irradiated with microwave energy immediately ahead of the mechanical excavation.
- *Microwave develops thermal stress in side the rock.
- *Rock is excavated with a lower force due to network of thermal generated fracture.
- *This can reduce 70% cutting force on drag bit.

Laser Technology I

- *Laser technology allows rapid kerfing
- *Mechanical excavator can break rock with the assistance of kerfing.
- *A thin kerf can be produced by laser relieving the confining force.
- *This allow the machine to cut the peripheral rock easily.
- *Laser beam can make thin deep cut by melting and vaporizing rock.

Laser Technology II

- *High energy laser penetrate rock with compressive strength up to 500Mpa.
- *Depth of penetration is linearly proportional to beam power.
- *Penetration depth is inversely proportional to scanning speed.
- *High power laser appear well suited as gage cutter for tunneling machines.
- *Application of laser shall reduce the machine down time due to rapid wear of gage cutter.

MI-1

MICROWAVE

TEST 1

MANG. HEM. CH. I-FM

20 KW

180 sec



0 10
Scale, cm

Thermal Fragmentation Application 1

- *It is possible to mine high grade narrow vein deposit of metal mines by this method .
- *This narrow vein can not be mined by conventional drilling & blasting due to high level of dilution.
- *Narrow vein of 50cm to 1m wide, between two sublevel drift can be mines.
- *This method is applied to high grade precious & base metal vein.

Thermal Fragmentation Theory 1

- *This mining is done by inserting a strong burner powered by diesel fuel & compressed air in to pilot hole previously drilled directly in to vein.
- *Series of 15cm (6inch) pilot holes are drilled in to vein with conventional drill.
- *Thermal fragmentation (thermal head, compressed air & water) is used to spall rock, quickly increasing hole diameter to 30 – 110cm in 10 minutes.
- *This extraction process allow thermal fragmentation with accuracy of 2cm.

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Thermal Fragmentation Method 1

- *Drift levels are excavated at 20m or 40m vertical interval in ore body.
- *Series of 15cm (6inch) pilot holes 20m long, are drilled in to vein (ore body) with conventional drill.
- *This holes can be drilled up ward or down ward, depending upon conditions.
- *Thermal head is inserted in to pilot hole.
- *Pilot hole is heated with help of diesel oil & compressed air along with water.

Thermal Fragmentation Method 2

- *Thermal reaction is created in side hole, spalling ore (rock) and hole is enlarged up to 1.1m.
- *Ore is broken to size below 13mm fragments.
- *In case hole is drilled downwards, broken ore is pulled up & loaded in to truck.
- *For up ward hole, broken ore gets directly loaded up on truck.
- *Remaining ore in between thermal holes is broken loose using low powered explosive, leaving waste wall intact.

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Thermal Fragmentation Method 3

*Fully mechanized equipment operated by 2 persons team (1 thermal fragmentation operator, 1 drill operator).

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Thermal Fragmentation Advantages 1

- *This method produces highly concentrated ore, resulting 400% to 500% less dilution, when compared to conventional mining method.
- * This mining method reduces environmental affect since smaller quantities or rock is disposed, stock piled & treated using chemical agents.
- *This method maximizes effectiveness of skilled personnel & increase productivity & safety.
- *This factor enhances profitability & resource utilization.

*

Thermal Fragmentation Advantages 2

- *There is no wall damage caused by blast vibration.
- *Since less rock is extracted, there is substantial cost reduction over entire chain of production.
- *Compact size of ore (0 to 13mm) & concentration of material brought to process plant increases production capacity of existing installation & output of product. Thermal unit is fully mechanized, reducing risk of physical injury.
- *Most of mine development is in ore.
- *This reduces generation of waste rock substantially.
- *Because of less dilution, chemical consumption for unit weight of metal is reduced.

Thermal Fragmentation Advantages 3

- *There is smaller tailing & waste pile in mines.
- *This method extends the life of mine.
- *Resource is converted to resources.
- **Drifting directly in to ore mining waste & development cost is reduced.
- *By using soft explosive charges, hang wall & foot wall are left intact & only mineralized corridor is extracted.
- *This fragmented rocks are fine & does not require crusher to process ore.
- *





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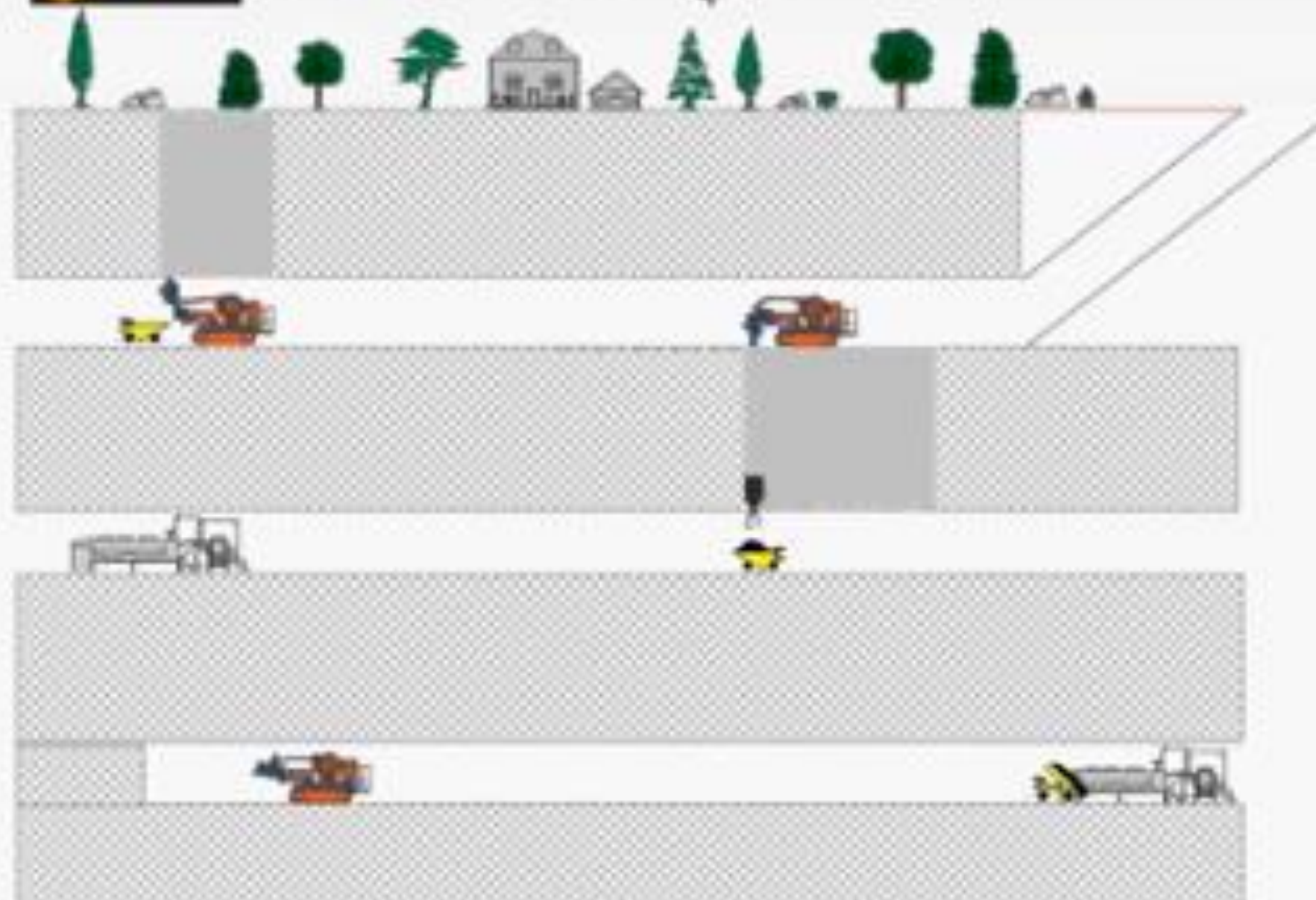
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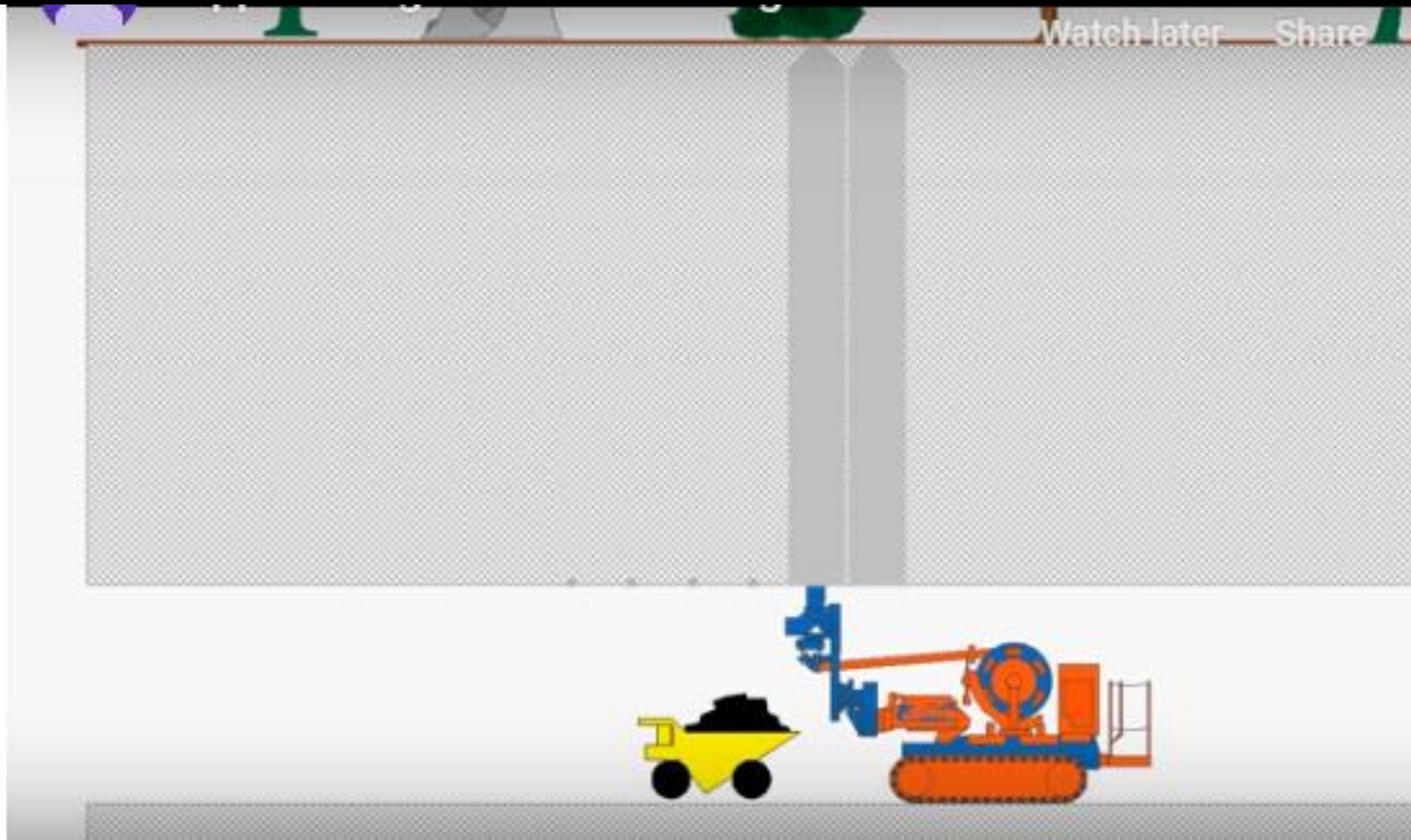
UNDERCUT GALLERY



TRELLISYS



Possible mining operations





11:11 PM - 11:11 PM

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Conventional Mining



Thermal Fragmentation

End result ;



If you have 20 g/t in a 50 cm vein,
you will have 20 g/t at the mill.
GOODBYE DILUTION!

Reference 1

*SME Mining Engineering Handbook, 2nd Edition
Volume 2, Howard L. Hartman. Page 2037 to 2040